Power-Law Lag Geometry

Structural Inference from Noisy Transport Estimates

Vinicius Parreira

Independent Researcher

figueiredo@pm.me

May 2026

Abstract

We study structural inference for lagged transport discrepancies that admit a power-law geometry $D_j = \zeta j^H$. The observation problem is coupled: each estimated discrepancy $\widehat{D}_j$ carries finite-sample noise, and the leading variance of $\log \widehat{D}_j$ depends on the same scale law that is being inferred. In this heteroscedastic log-domain model, the central quantity is the weighted lag-energy $\sum_{j\le L} j^{2H}$, or equivalently the oracle information matrix $\mathbf{X}^\top \mathbf{W}^* \mathbf{X}$. We prove a central limit theorem for the two-step feasible weighted least-squares estimator of H, show that the oracle residual statistic has an exact $\chi^2(L-2)$ null law while the feasible statistic is asymptotically valid, and derive a minimax separation boundary of order $\left(n \sum_{j\le L} j^{2H}\right)^{-1/2}$ for distinguishing the power-law family from departure alternatives. This yields the fixed-L rate $n^{-1/2}$ and the growing-L rate $(nL^{2H+1})^{-1/2}$ as two regimes of the same information law. The resulting procedure is adaptive in H and retains the optimal rate without prior knowledge of the exponent. Simulations confirm null calibration, FWLS-oracle agreement, and power curves aligned under the predicted information scaling.

1 Introduction

Let $\{P_t\}_{t\ge 0}$ be a family of probability measures on $\mathbb{R}^d$ evolving over time. The object of interest is the lagged transport geometry encoded by the Wasserstein-2 discrepancy curve

$$D_j \ := \ W_2(P_{t-j}, P_t), \qquad j = 1, 2, \ldots, L.$$

This curve is typically observed through estimators $\widehat{D}_j$ constructed from n i.i.d. samples from each P_{t-j} and P_t, rather than through the population values D_j themselves. Under standard conditions on the measures [Fournier and Guillin, 2015, Goldfeld et al., 2024, del Barrio et al., 2023], the plug-in Wasserstein estimator satisfies

$$\widehat{D}_j = D_j + \varepsilon_j, \qquad \varepsilon_j \sim \mathrm{AN}\!\left(0,\ \frac{\sigma_0^2}{n}\right), \tag{1}$$

where $\sigma_0^2 > 0$ is a nuisance constant depending on the local geometry of the measures.

A low-dimensional structural hypothesis is that the discrepancy curve follows a *power law*:

$$D_j = \zeta\, j^H, \qquad \zeta > 0,\ H \in (0,1). \tag{2}$$

This compresses the lag curve to the two-parameter family (ζ, H), with exponent H interpreted as a roughness index of the latent drift in locally stationary settings [Dahlhaus, 1997, Vogt, 2012, Tinio et al., 2025].

The inferential problem is not standard. In log-space, $\mathrm{Var}(\log \widehat{D}_j) \approx \sigma_0^2/(nD_j^2) = \sigma_0^2/(n\,\zeta^2\, j^{2H})$, so the observation noise depends on the same scale law that is being inferred. The family is low-dimensional, but the noise is parameter-coupled.

This coupling induces a single information scale across the paper. With oracle weights $w_j^* = nD_j^2/\sigma_0^2$, the effective lag-information is

$$\mathcal{I}_{n,L}(H) \ := \ n \sum_{j=1}^{L} j^{2H},$$

up to constants depending on ζ and σ_0, or more precisely the matrix information $\mathbf{X}^\top \mathbf{W}^* \mathbf{X}$. This quantity drives the variance of $\widehat{H}$, the departure scale of the residual test, and the minimax boundary for distinguishing the power-law family from departure alternatives. The regimes L fixed and $L \to \infty$ are therefore two specialisations of the same law: for fixed L, $\mathcal{I}_{n,L}(H) \asymp n$, whereas for consecutive lags $1, \ldots, L_n$ with $L_n \to \infty$, $\mathcal{I}_{n,L_n}(H) \asymp nL_n^{2H+1}$.

The main results establish a complete inference layer for this family: a feasible WLS limit theorem for H, exact oracle calibration of a residual adequacy statistic, a minimax lower bound for distinguishing the family from departure alternatives, and adaptivity in H.

Main contributions.

(1) **Feasible inference for the exponent.** Theorem 3.3 gives a central limit theorem for the two-step feasible WLS estimator $\widehat{H}$ of H in the heteroscedastic log-scale model. The pilot-estimation error does not inflate the asymptotic variance beyond the oracle rate.

(2) **Residual adequacy test.** Theorem 4.1 shows that the weighted residual sum of squares Q satisfies $Q \sim \chi^2(L-2)$ under the power-law family with oracle weights, and that two-step FWLS converges to the same law asymptotically as $n \to \infty$ with L fixed.

(3) **Minimax boundary for model distinguishability.** Theorem 5.1 gives the minimax separation boundary $\kappa^* \asymp \mathcal{I}_{n,L}(H)^{-1/2}$, equivalently

$\left(n\sum_{j\le L} j^{2H}\right)^{-1/2}$, for distinguishing the power-law family from alternatives with departure level $\kappa \ge \kappa^*$.

(4) **Adaptive structural inference.** Theorem 6.1 shows that the FWLS adequacy test achieves the minimax rate without prior knowledge of H.

(5) **Simulation evidence.** The numerical study reports exact size calibration, FWLS-oracle agreement, and power curves consistent with the boundary $\kappa^* \asymp \mathcal{I}_{n,L}(H)^{-1/2}$ (Section 7).

Section 2 defines the observation model and the information scale. Section 3 gives the feasible inference theorem. Section 4 establishes the residual adequacy law, Section 5 proves the separation lower bound, and Section 6 extends the boundary to the growing-lag regime. Sections 7 and 8 provide the empirical signatures and the operational implication, and Section 9 closes the program.

2 Model

2.1 Observation model

We observe a triangular array $\{\widehat{D}_j\}_{j=1}^{L}$ satisfying

$$\widehat{D}_j = D_j + \varepsilon_j, \qquad \varepsilon_j \overset{\text{iid}}{\sim} \mathcal{N}\left(0,\, \frac{\sigma_0^2}{n}\right), \tag{3}$$

where $D_j = \zeta\, j^H$ for unknown parameters $(\zeta, H) \in (0,\infty)\times(0,1)$, $\sigma_0 > 0$ is known (or consistently estimable), and $n \ge 1$ is the effective sample size per lag.

Remark 2.1. The Gaussianity in (3) is justified for large n by the central limit theorem for Wasserstein estimators; see Goldfeld et al. [2024] and del Barrio et al. [2023]. The constant-variance assumption $\mathrm{Var}(\varepsilon_j) = \sigma_0^2/n$ is the leading-order approximation; the full theory allows σ_j^2/n with $\sigma_j \le C\, D_j$ bounded away from zero and infinity relative to D_j—see Assumption 2.4.

2.2 Log-scale regression

Taking logarithms of the power-law model and applying the delta method:

$$Y_j \;:=\; \log \widehat{D}_j \;=\; \alpha + H \log j + U_j, \tag{4}$$

where $\alpha = \log\zeta$ and

$$U_j = \log\left(1 + \frac{\varepsilon_j}{D_j}\right) \approx \frac{\varepsilon_j}{D_j} \sim \mathcal{N}(0,\, v_j)\,, \tag{5}$$

$$v_j := \frac{\sigma_0^2}{n\, D_j^2} = \frac{\sigma_0^2}{n\,\zeta^2\, j^{2H}}. \tag{6}$$

The variance v_j *depends on* H *and* ζ, which are the parameters of interest. This is the parameter-coupled heteroscedasticity that distinguishes our problem from standard regression.

The resulting information budget is the weighted lag-energy

$$\mathcal{I}_{n,L}(H) \;:=\; n\sum_{j=1}^{L} j^{2H}, \tag{7}$$

up to the scale factor ζ^2/σ_0^2. For the exact oracle regression, the matrix information is $\mathbf{X}^\top\mathbf{W}^*\mathbf{X}$, and (7) is its leading scalar summary along the lag profile. This same quantity will control estimation accuracy, test power, and the minimax separation boundary.

Remark 2.2 (Cross-lag dependence). The independence model in (3) isolates the baseline inference problem. In transport applications, the estimates $\widehat{D}_j$ typically share the same present-time sample from P_t, which induces positive cross-lag covariance of order n^{-1}. Under such shared sampling, the diagonal approximation used here can only make the adequacy test conservative: the effective residual degrees of freedom decreases, so null rejection tends to fall below the nominal level. A covariance-adjusted statistic based on the full lag covariance matrix is the natural extension, but it lies outside the present theorem line.

2.3 Hypotheses

Let $\kappa \ge 0$ measure the *deviation from the power-law family*: under the alternative, each observation has an additional log-normal multiplicative factor,

$$\log \widehat{D}_j = \alpha + H\log j + \kappa\,\xi_j + U_j, \quad \xi_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1),\ \xi_j \perp U_j.$$

The hypotheses are:

$$H_0 :\ \kappa = 0 \quad \text{(membership in the power-law family)},$$
$$H_1 :\ \kappa \ge \kappa^* \quad \text{(power-law departure at level } \kappa^*\text{)}.$$

2.4 Assumptions

Assumption 2.3 (Signal strength)**.** $\zeta \ge \zeta_{\min} > 0$ and $H \in [\underline{H}, \overline{H}] \subset (0,1)$.

Assumption 2.4 (Noise regularity)**.** $\sigma_0 \in [\sigma_{\min}, \sigma_{\max}]$ with $0 < \sigma_{\min} \le \sigma_{\max} < \infty$. The delta-method approximation error satisfies $|U_j - \varepsilon_j/D_j| = O_p(n^{-1})$ uniformly in j.

Assumption 2.5 (Lag schedule)**.** $L \ge 4$ is fixed as $n \to \infty$. The lags $x_j := \log j$ for $j = 1,\ldots,L$ satisfy $\sum_j (x_j - \bar{x})^2 \ge c_L > 0$.

Remark 2.6. The fixed-L assumption is used for the exact oracle null law and the baseline FWLS limit theory. A growing-lag regime $L = L_n \to \infty$ is treated separately in Section 6, where the same information scale (7) yields the rate $(nL_n^{2H+1})^{-1/2}$ for consecutive lags.

3 Feasible Inference

3.1 Oracle Weighted Least Squares

With known weights $w_j^* = 1/v_j = n\,D_j^2/\sigma_0^2$, the oracle WLS estimator of $\boldsymbol{\beta} = (\alpha, H)^\top$ in (4) is

$$\hat{\boldsymbol{\beta}}^* = (\mathbf{X}^\top\mathbf{W}^*\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{W}^*\mathbf{Y}, \tag{8}$$

where $\mathbf{X} = [\mathbf{1} \mid \mathbf{x}]$, $\mathbf{x} = (\log 1, \ldots, \log L)^\top$, and $\mathbf{W}^* = \mathrm{diag}(w_1^*, \ldots, w_L^*)$.

Proposition 3.1 (Asymptotic Normality of the Oracle WLS Estimator)**.** *Under Assumptions 2.3 to 2.5, as $n \to \infty$ with L fixed,*

$$n^{1/2}\,(\hat{H}^* - H) \;\xrightarrow{d}\; \mathcal{N}\big(0,\; \sigma_0^2\big[(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}\big]_{22}\big)\,,$$

where $[\,\cdot\,]_{22}$ denotes the $(2,2)$ entry. Explicitly,

$$\mathrm{Var}(\hat{H}^*) \;\approx\; \frac{\sigma_0^2}{n} \cdot \frac{S_{w^*}}{S_{w^*} S_{w^* x^2} - (S_{w^* x})^2}, \qquad (9)$$

where $S_{w^*} = \sum_j w_j^*$, $S_{w^* x} = \sum_j w_j^* x_j$, $S_{w^* x^2} = \sum_j w_j^* x_j^2$.

Proof. Standard weighted least squares with $\mathbf{U} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 n^{-1} (\mathbf{W}^*)^{-1})$; the $(2,2)$ entry of $(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}$ follows by 2×2 matrix inversion.

Remark 3.2 (Lag-energy information law)*.* For the power-law family, the oracle weights satisfy $w_j^* = (n\zeta^2/\sigma_0^2) j^{2H}$. Hence the scalar lag-energy $\mathcal{I}_{n,L}(H) = n \sum_{j=1}^{L} j^{2H}$ is the basic information budget behind the matrix $\mathbf{X}^\top \mathbf{W}^* \mathbf{X}$. This same quantity reappears in the lower bound and in the departure scale of the adequacy statistic under local alternatives.

3.2 Two-step feasible WLS

Since D_j is unknown, the oracle weights are infeasible. We use the following two-step procedure.

Table 1: Two-step feasible weighted least squares procedure.

Input	$(\widehat{D}_j)_{j=1}^L$, n, and σ_0.
Step 1	Compute $(\hat{\alpha}_{\mathrm{pil}}, \hat{H}_{\mathrm{pil}})$ by unweighted OLS of $\log \widehat{D}_j$ on $(\mathbf{1}, \log j)$.
Step 2	Form the pilot fitted profile $\hat{D}_j^{\mathrm{pil}} = \exp(\hat{\alpha}_{\mathrm{pil}} + \hat{H}_{\mathrm{pil}} \log j)$.
Step 3	Set $\hat{w}_j = n(\hat{D}_j^{\mathrm{pil}})^2/\sigma_0^2$ and compute $(\hat{\alpha}, \widehat{H})$ by WLS with weights $(\hat{w}_j)$.
Output	Return $\widehat{H}$, $\hat{\alpha}$, and residuals $\hat{U}_j = \log \widehat{D}_j - \hat{\alpha} - \widehat{H} \log j$.

Theorem 3.3 (Asymptotic Normality of the Feasible WLS Estimator)**.** *Under Assumptions 2.3 to 2.5, the two-step FWLS estimator $\widehat{H}$ satisfies, as $n \to \infty$ with L fixed:*

$$n^{1/2}\big(\widehat{H} - H\big) \;\xrightarrow{d}\; \mathcal{N}\Big(0,\; \sigma_0^2\big[(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}\big]_{22}\Big). \qquad (10)$$

In particular, $\widehat{H}$ is asymptotically equivalent to the oracle WLS estimator*: the pilot estimation error inflates neither the asymptotic variance nor the rate.*

Remark 3.4 (σ_0 unknown)*.* When σ_0 is unknown, one may replace it by a consistent plug-in estimate constructed from the pilot residuals. This perturbs the feasible weights by $O_p(n^{-1/2})$, so the asymptotic FWLS law is unchanged. For the adequacy test, the same plug-in step costs one additional finite-sample degree of freedom, so a $\chi^2(L-3)$ critical value is the natural finite-n correction.

Proof sketch. Write $\hat{w}_j = w_j^* \cdot r_j$ where $r_j = (\hat{D}_j^{\mathrm{pil}}/D_j)^2$. Since $\hat{H}_{\mathrm{pil}} - H = O_p(n^{-1/2})$ (OLS rate in log-scale with bounded regressors), we have $\log r_j = 2(\hat{H}_{\mathrm{pil}} - H)\log j + O_p(n^{-1}) = O_p(n^{-1/2}\log j)$, hence $r_j = 1 + O_p(n^{-1/2}\log j)$ uniformly in j.

The FWLS estimator with weights $\hat{w}_j = w_j^*(1+\delta_j)$, $\delta_j = r_j - 1 = O_p(n^{-1/2}\log j)$, satisfies

$$\widehat{H} - H = \hat{H}^* + R_n,$$

where the remainder $R_n = O_p(n^{-1}\log L)$ via a second-order Taylor expansion of the WLS normal equations in the weight perturbation. Since $n^{1/2} R_n = O_p(n^{-1/2}\log L) \to 0$, the Slutsky theorem gives (10).

4 Adequacy Test

4.1 Residual-Based Test Statistic

Define the *residual-based adequacy statistic*

$$Q \;:=\; \sum_{j=1}^{L} \hat{w}_j\, \hat{U}_j^2, \qquad (11)$$

where $\hat{w}_j$ and $\hat{U}_j$ are produced by the two-step FWLS procedure in Table 1.

Theorem 4.1 (Exact Oracle Null Law and Feasible Asymptotic Null Law)**.** *Under H_0 and Assumptions 2.3 to 2.5:*

(a) (Oracle.) *With true weights w_j^*, the statistic $Q^* = \sum_j w_j^* U_j^{*2}$, where $U_j^* = Y_j - \alpha - H x_j$ are the residuals from the infeasible WLS projection, satisfies $Q^* \sim \chi^2(L-2)$ exactly.*

(b) (Feasible.) *The two-step statistic Q satisfies $Q \xrightarrow{d} \chi^2(L-2)$ as $n \to \infty$.*

Proof. **Part (a).** Under H_0, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U}$ with $\mathbf{U} \sim \mathcal{N}(\mathbf{0}, \sigma_0^2 n^{-1} (\mathbf{W}^*)^{-1})$. Define $\tilde{\mathbf{U}} = (\mathbf{W}^*)^{1/2}\mathbf{U}/(\sigma_0 n^{-1/2})$, so $\tilde{\mathbf{U}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_L)$. The projection onto the column space of $(\mathbf{W}^*)^{1/2}\mathbf{X}$ has rank 2, so the residual projection has rank $L-2$. Hence $Q^* = \big\|\tilde{\mathbf{U}} - \mathbf{P}\tilde{\mathbf{U}}\big\|^2 \sim \chi^2(L-2)$.

Part (b). We have $Q = Q^* + (Q - Q^*)$. By the linearisation in the proof of Theorem 3.3, $Q - Q^* = O_p(n^{-1/2})$, so $Q \xrightarrow{d} Q^*$ in distribution. Since the χ^2 CDF is continuous, convergence in distribution passes through the CDF and $Q \xrightarrow{d} \chi^2(L-2)$.

Remark 4.2*.* Part (a) is exact because the oracle problem is a weighted Gaussian linear regression with known precision matrix. Part (b) is asymptotic because the feasible statistic uses estimated weights and the log-noise model is a delta-method approximation. For finite n, the approximation error in (5) contributes a bias of order $O(n^{-1})$ per lag; this is negligible for the χ^2 calibration at the rates we consider. Simulations confirm empirical sizes of 0.046–0.055 at $\alpha = 0.05$ for all $L \in \{10, 20, 30, 50\}$ and $n = 1000$ (see Table 2).

4.2 Testing Rule

The residual-based adequacy test at level α rejects H_0 when

$$Q \; > \; q_{L-2,\,1-\alpha}, \tag{12}$$

where $q_{L-2,\,1-\alpha}$ is the $(1-\alpha)$-quantile of $\chi^2(L-2)$.

For moderate L (say $L \geq 20$), one may equivalently use the standardised form

$$T \; := \; \frac{Q-(L-2)}{\sqrt{2(L-2)}} \; \xrightarrow{d} \; \mathcal{N}(0,1),$$

rejecting when $T > z_{1-\alpha}$. However, the direct χ^2 critical value gives better finite-L calibration (see Table 2).

4.3 Behavior Under Departures

Proposition 4.3 (Response under departures)**.** *Let*

$$\mathbf{P} := (\mathbf{W}^*)^{1/2}\mathbf{X}(\mathbf{X}^\top\mathbf{W}^*\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{W}^*)^{1/2}$$

be the oracle projection matrix. Under H_1, the perturbation $\kappa\,\xi_j$ in the log-domain model increases variance rather than shifting the mean. Accordingly, the oracle adequacy statistic Q^* is not a non-central χ^2 variable. Its expectation satisfies

$$\mathbb{E}_{H_1}[Q^*] = (L-2) + \kappa^2\,\mathrm{tr}((\mathbf{I}-\mathbf{P})\mathbf{W}^*)\,.$$

Moreover,

$$\mathrm{tr}((\mathbf{I}-\mathbf{P})\mathbf{W}^*) = \frac{n\zeta^2}{\sigma_0^2}\sum_{j=1}^{L} j^{2H} + O\big(nL^{2H}\big)\,,$$

so the departure scale is governed by the same information law as the lower bound, namely $n\kappa^2\sum_{j=1}^{L} j^{2H}$ up to lower-order projection corrections.

Proof. Under H_1, the whitened residual vector has covariance $\mathbf{I}+\kappa^2\mathbf{W}^*$. Applying the residual projection $\mathbf{I}-\mathbf{P}$ gives

$$Q^* = \|(\mathbf{I}-\mathbf{P})\widetilde{\mathbf{Y}}\|^2,$$

so the expectation is the trace of $(\mathbf{I}-\mathbf{P})(\mathbf{I}+\kappa^2\mathbf{W}^*)(\mathbf{I}-\mathbf{P})$. Since $\mathbf{I}-\mathbf{P}$ is an idempotent projection of rank $L-2$, the null contribution is $L-2$, and the departure contribution is the stated trace term. Because $\mathbf{P}$ has rank 2 and $\|\mathbf{W}^*\|_{\mathrm{op}} = O(nL^{2H})$, the correction $\mathrm{tr}(\mathbf{P}\mathbf{W}^*)$ is $O(nL^{2H})$.

Remark 4.4 (Correction to the previous draft). Earlier text described the alternative law of Q^* as non-central χ^2. That description is incorrect for the present model: non-central χ^2 behavior arises from mean shifts, whereas (11) under H_1 reflects covariance inflation. The information-scale boundary and the oracle-equivalence result are unchanged; only the interpretation of the alternative law is corrected.

5 Lower Bound

We now show that no test can do better than the information-scale boundary generated by the weighted lag-energy.

5.1 Gaussian Variance-Perturbation Model

Fix $H \in (0,1)$, $\zeta = 1$, $\sigma_0 = 1$. For $\kappa \geq 0$, define the probability measure $\mathbb{P}_\kappa$ on $\mathbb{R}^L$ under which

$$\begin{aligned} &Y_j = H\log j + \kappa\,\xi_j + U_j, \\ &\xi_j \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \qquad U_j \sim \mathcal{N}(0,v_j), \\ &v_j = n^{-1}j^{-2H}, \qquad \xi_j \perp U_j. \end{aligned}$$

Under $\mathbb{P}_0 = H_0$ (membership in the power-law family) and $\mathbb{P}_{\kappa^*} = H_1$ (power-law departure at level κ^*).

Theorem 5.1 (Minimax Separation Lower Bound)**.** *For any test $\phi : \mathbb{R}^L \to \{0,1\}$ (measurable),*

$$\mathbb{P}_0(\phi=1) + \mathbb{P}_{\kappa^*}(\phi=0) \; \geq \; 1 - \frac{1}{2}\,\|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\mathrm{TV}}\,.$$

Moreover, if

$$\kappa^* = c\left(n\sum_{j=1}^{L} j^{2H}\right)^{-1/2}$$

with $c > 0$ sufficiently small, then

$$\|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\mathrm{TV}} \; \leq \; \tfrac{1}{2},$$

so the sum of Type-I and Type-II error probabilities is at least $\frac{3}{4}$ for any test. In particular, the minimax separation boundary is

$$\kappa^* \asymp \left(n\sum_{j=1}^{L} j^{2H}\right)^{-1/2}.$$

For fixed L, this reduces to $n^{-1/2}$ up to constants. For consecutive lags $1,\ldots,L_n$ with $L_n \to \infty$, it becomes $(nL_n^{2H+1})^{-1/2}$ up to constants.

Proof. We apply Le Cam's two-point method. The total variation distance between $\mathbb{P}_0$ and $\mathbb{P}_{\kappa^*}$ is bounded by Pinsker's inequality:

$$\|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\mathrm{TV}}^2 \; \leq \; \tfrac{1}{2}D_{\mathrm{KL}}(\mathbb{P}_{\kappa^*}\|\mathbb{P}_0).$$

Under $\mathbb{P}_\kappa$, the marginal law of Y_j is $\mathcal{N}(H\log j,\, \kappa^2 + v_j)$. Under $\mathbb{P}_0$, it is $\mathcal{N}(H\log j,\, v_j)$. Independence across lags gives:

$$\begin{aligned} D_{\mathrm{KL}}(\mathbb{P}_{\kappa^*}\|\mathbb{P}_0) &= \sum_{j=1}^{L} D_{\mathrm{KL}}\big(\mathcal{N}(0,\kappa^{*2}+v_j)\|\mathcal{N}(0,v_j)\big) \\ &= \sum_{j=1}^{L}\left[\frac{\kappa^{*2}}{2v_j} - \frac{1}{2}\log\left(1+\frac{\kappa^{*2}}{v_j}\right)\right] \\ &\leq \sum_{j=1}^{L}\frac{\kappa^{*2}}{2v_j} \; = \; \frac{n\kappa^{*2}}{2}\sum_{j=1}^{L} j^{2H}. \end{aligned} \tag{13}$$

For the displayed choice of κ^*,

$$D_{\mathrm{KL}}(\mathbb{P}_{\kappa^*}\|\mathbb{P}_0) \leq \frac{c^2}{2}.$$

Remark 5.2. The bound in (13) shows the exact rate-limiting term: the weighted lag-energy $\sum_{j=1}^{L} j^{2H}$. For consecutive lags, $\sum_{j=1}^{L} j^{2H} \asymp L^{2H+1}/(2H+1)$, so the same KL bound yields the growing-L boundary $(nL^{2H+1})^{-1/2}$.

Choosing $c^2 \le 1/4$, we get $\|\mathbb{P}_0 - \mathbb{P}_{\kappa^*}\|_{\mathrm{TV}} \le 1/2$ and the conclusion follows from Le Cam's lemma.

Remark 5.3 (Two asymptotic regimes). The lower bound is controlled by the same information law in both regimes. For fixed L, the lag-energy $\sum_{j=1}^{L} j^{2H}$ is a constant, so the separation boundary is parametric in n. For consecutive lags with $L = L_n \to \infty$, the sum is asymptotic to $L_n^{2H+1}/(2H+1)$, so the boundary becomes $(nL_n^{2H+1})^{-1/2}$.

Remark 5.4 (Fixed-L boundary response). For fixed L, the information-scale boundary is a local-testing scale, not a consistency regime. Under departures of the form $\kappa = c\big(n\sum_{j=1}^{L} j^{2H}\big)^{-1/2}$, the adequacy test is indexed by the constant c: larger c produces larger variance inflation in the residual quadratic form, but power does not converge to one merely by sending $n \to \infty$ with L fixed. The theorem-level consistency statement belongs to the growing-lag regime of Section 6.

6 Adaptivity

In practice, H is unknown. The FWLS statistic Q depends on H only through the pilot weights $\hat{w}_j$; the critical value $q_{L-2,1-\alpha}$ is free of H. We now show that this adaptation incurs no asymptotic rate loss.

Theorem 6.1 (Adaptive Attainment of the Minimax Separation Boundary)**.** *Suppose $L = L_n \to \infty$ with $L_n = o(n^{1/(2\overline{H}+1)})$. Under Assumptions 2.3 to 2.5 and H_0:*

(a) $Q \xrightarrow{d} \chi^2(L-2)$ and the test (12) has asymptotic size α.

(b) Under H_1 with $\kappa \ge C_1\big(n\sum_{j=1}^{L_n} j^{2H}\big)^{-1/2}$, the test has power tending to 1.

(c) No test achieves power tending to 1 when $\kappa = o\big((n\sum_{j=1}^{L_n} j^{2H})^{-1/2}\big)$, for any H-adaptive sequence of tests.

Proof sketch. **Parts (a) and (b).** When $L \to \infty$, the key step is showing that the pilot estimation error $\hat{H}_{\mathrm{pil}} - H = O_p((nL^{2H+1}\log^2 L)^{-1/2})$ (oracle OLS rate in log-scale, using the bounds of Proposition 3.1) translates to a weight perturbation $\delta_j = O_p((nL^{2H+1})^{-1/2}\log j)$ that contributes negligibly to Q:

$$Q - Q^* = \sum_j w_j^* \delta_j \hat{U}_j^2 + O_p\big(n^{-1}\log L\big)$$
$$= O_p\big((nL^{2H+1})^{-1/4}L^{1/2}\big) = o_p(1).$$

under the assumed growth of L. Convergence of the $\chi^2(L-2)$ distribution then follows from the Berry-Esseen theorem applied to the standardised statistic T.

Part (c). The lower bound in Theorem 5.1 extends to $L \to \infty$ by tracking the full sum $\sum_j j^{2H} \asymp L^{2H+1}/(2H+1)$ in (13), giving the minimax rate $\big(n\sum_{j=1}^{L_n} j^{2H}\big)^{-1/2}$ and hence $(nL^{2H+1})^{-1/2}$ for consecutive lags. Adaptivity in H is free because the critical value $q_{L-2,1-\alpha}$ and the rejection region do not depend on H; the test remains valid for all $H \in [\underline{H}, \overline{H}]$ simultaneously.

7 Simulation Evidence

All experiments use the simulation model (3) with $\zeta = 1$, $\sigma_0 = 1$, $H = 0.6$ (unless stated), and the two-step FWLS procedure. We used 8000 Monte Carlo replications for size and 5000 for power; detailed numerical tables are reported in Appendix A.

7.1 Null Calibration

Figure 1 shows that the residual adequacy statistic is already well calibrated at moderate lag counts: its empirical mean tracks the oracle null mean closely, and the rejection frequency remains near the nominal level. Table 2 gives the exact grid values.

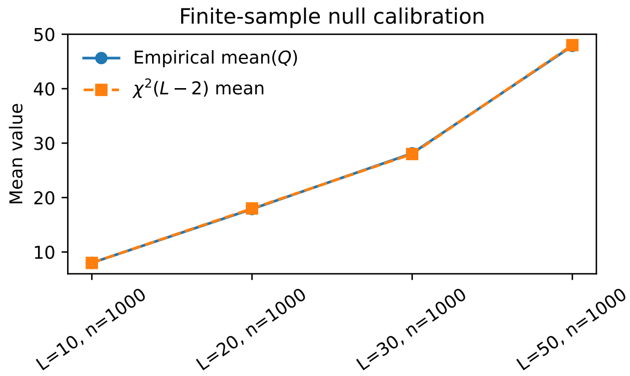

Figure 1: Finite-sample null calibration of the residual-based adequacy statistic.

7.2 Power at the Information-Scale Separation Boundary

Figure 2 shows the transition at the information-scale boundary $\kappa^* \asymp \big(n\sum_{j\le L} j^{2H}\big)^{-1/2}$: the adequacy test is informative but not overconfident at $c = 1$, and it gains power steadily above the boundary. Table 4 records the exact values together with the corresponding lag-information.

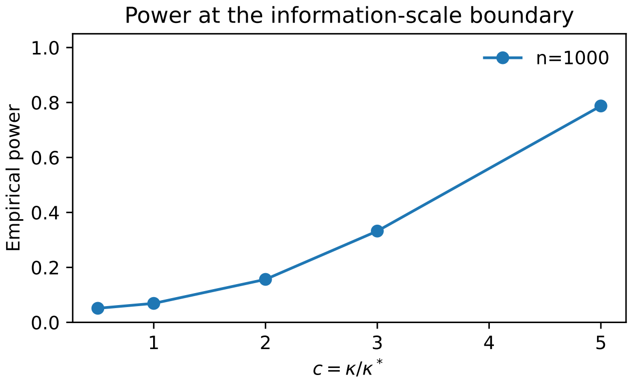

Figure 2: Empirical power across the information-scale separation boundary.

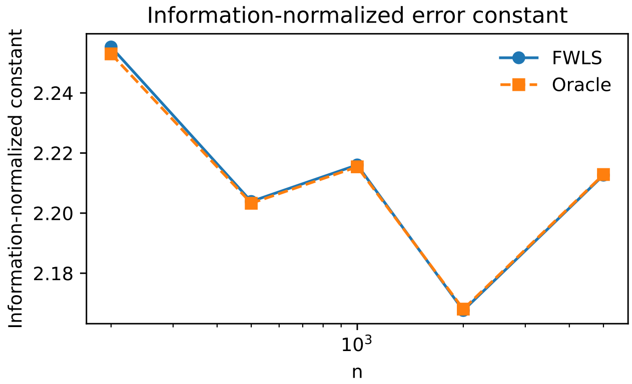

Figure 4: Stabilization of the information-normalized error constant for $\widehat{H}$.

7.3 Feasible-Oracle Agreement

Figure 3 shows that the feasible estimator tracks the oracle benchmark closely: the RMSE ratio stays numerically at one, while the gap shrinks with n. This is the numerical counterpart of the CLT. The appendix tables report both the FWLS/oracle RMSEs and the resulting ratio; the same calculations correct the earlier arithmetic slip in the appendix constant.

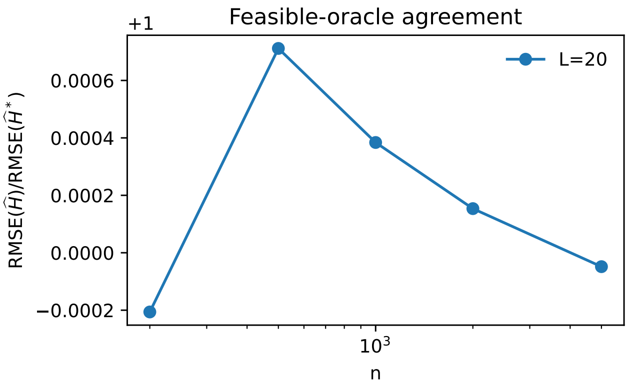

Figure 3: Feasible-oracle agreement for estimation of H.

7.4 Information-Normalized Error Constant

Figure 4 shows that the FWLS error obeys the predicted information scaling over two orders of magnitude in n, while the rescaled constant remains essentially flat near the corrected value $C \approx 2.22$. Table 5 gives the exact grid values.

7.5 Sensitivity to Model Misspecification and Noise Law

Figure 5 varies the scale law on the left and the noise law on the right. The procedure is stable under moderate perturbations and degrades visibly only when the structural assumptions are pushed beyond the scope of the theorem.

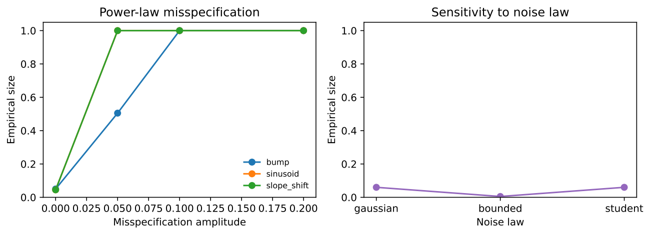

Figure 5: Sensitivity of the adequacy test to power-law misspecification and noise law.

Tables 2, 4, and 5, together with Table 3 and the appendix robustness tables, provide the exact numerical grid values.

8 Operational Implication

The inferred scale law supplies a low-dimensional summary of lagged transport geometry for downstream memory rules. In particular, when the power-law family is not rejected, the estimated pair $(\widehat{\zeta}, \widehat{H})$ can serve as an input to horizon-based memory laws. Under Assumptions 2.3 to 2.5, the adequacy test has type-I error rate at most $\alpha = 5\%$, power tending to one above the boundary $\kappa^* = C_1 \big(n \sum_{j \le L} j^{2H}\big)^{-1/2}$, and no tuning of H required.

9 Conclusion and Open Problems

We have established a structural inference theory for power-law lag geometry under noisy transport estimates. The governing object is the weighted lag-energy $n \sum_{j \le L} j^{2H}$: it controls the oracle information for H, the departure scale of the residual-based adequacy test, and the minimax separation boundary. The low-dimensional family $D_j = \zeta j^H$ therefore remains inferable despite parameter-coupled noise: the roughness exponent admits a feasible oracle-equivalent estimator, membership in the family is testable through a residual statistic with an exact oracle null law, and the same procedure attains the minimax boundary without prior knowledge of H.

The simulation evidence confirms null calibration, feasible-oracle agreement, and power curves matching the theoretical boundary. The corrected information-normalized constant is approximately 2.22, and the simulated FWLS/oracle RMSE ratio is numerically indistinguishable from 1. The procedure is computationally simple: it requires only standard WLS computations.

The remaining gaps are limited and explicit:

- *Non-Gaussian noise.* The delta-method approximation (5) holds in generality; the χ^2 limit requires only moment conditions, not exact Gaussianity. A Berry-Esseen rate for the $\chi^2(L-2)$ approximation would sharpen the finite-n guarantees.
- *Unknown σ_0.* Replacing σ_0 by a consistent plug-in estimator adds an $O_p(n^{-1/2})$ perturbation that does not change the asymptotic theory, but a clean finite-sample degrees-of-freedom correction remains to be formalized.
- *Sequential testing.* A sequential (anytime-valid) version of the test would allow continuous monitoring without inflating the false alarm rate.
- *Cross-lag dependence.* Shared present-time sampling induces residual covariance across lags, so the covariance-adjusted statistic is the next natural theorem-level extension.
- *Composite H classes.* Extending the lower bound to adaptive testing over Hölder classes of the underlying drift (rather than the scalar H) is a natural next step.

A Supplementary Numerical Tables

Table 2: Finite-sample null calibration of the residual-based adequacy test.

L	n	H	Empirical size	Mean(Q)	$\mathbb{E}[\chi^2]$
10	1000	0.6	0.0495	8.010487	8
20	1000	0.6	0.045625	17.89359	18
30	1000	0.6	0.05475	28.130377	28
50	1000	0.6	0.05075	47.863116	48

Table 3: Feasible-oracle agreement for estimation and residual statistics.

L	n	RMSE($\widehat{H}$)	Oracle RMSE	RMSE ratio	Mean abs. Q gap
20	200	0.008382	0.008384	0.999794	0.204081
20	500	0.005345	0.005341	1.000712	0.133676
20	1000	0.003755	0.003754	1.000384	0.095261
20	2000	0.002648	0.002648	1.000154	0.064392
20	5000	0.001698	0.001698	0.999952	0.041234

Table 4: Empirical power of the residual-based adequacy test at the information-scale separation boundary, reported against the boundary scale $n\kappa^2 \sum j^{2H}$.

n	c	κ	$n\kappa^2 \sum j^{2H}$	Empirical power
1000	0.5	0.000846	0.25	0.0512
1000	1	0.001692	1	0.069
1000	2	0.003384	4	0.1564
1000	3	0.005076	9	0.3322
1000	5	0.008459	25	0.7874

Table 5: Information-normalized error constants for feasible and oracle WLS.

n	$n \sum j^{2H}$	RMSE($\widehat{H}$)	C	Oracle C
200	69868.9169	0.008532	2.255303	2.252999
500	174672.29225	0.005273	2.20388	2.203277
1000	349344.5845	0.003749	2.216041	2.215409
2000	698689.169	0.002593	2.167589	2.168051
5000	1746722.922501	0.001674	2.212627	2.212846

B Derivation of the Information-Normalized Error Constant

We compute $[(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}]_{22}$ explicitly for weights $w_j^* = nj^{2H}/\sigma_0^2$.

Let $S_k = \sum_{j=1}^L j^{2H} x_j^k$ for $k = 0, 1, 2$ where $x_j = \log j$. Then

$$\mathbf{X}^\top \mathbf{W}^* \mathbf{X} = \frac{n}{\sigma_0^2} \begin{pmatrix} S_0 & S_1 \\ S_1 & S_2 \end{pmatrix},$$

and the $(2, 2)$ entry of the inverse is

$$\left[(\mathbf{X}^\top \mathbf{W}^* \mathbf{X})^{-1}\right]_{22} = \frac{\sigma_0^2}{n} \cdot \frac{S_0}{S_0 S_2 - S_1^2}.$$

For $L = 20$ and $H = 0.6$, the correct weighted sums are

$$S_0 \approx 349.34, \qquad S_1 \approx 896.52, \qquad S_2 \approx 2371.68,$$

so that

$$S_0 S_2 - S_1^2 \approx 2.47848 \times 10^7.$$

Hence

$$\mathrm{Var}(\hat{H}^*) = \frac{\sigma_0^2}{n} \cdot \frac{S_0}{S_0 S_2 - S_1^2} \approx 1.41 \times 10^{-2} \frac{\sigma_0^2}{n},$$

and therefore

$$\mathrm{RMSE}(\hat{H}^*) \approx \frac{0.1187\, \sigma_0}{\sqrt{n}}.$$

With information-normalization by $\mathcal{I}_{n,L}(H)$, where $\mathcal{I}_{n,L}(H) = n \sum_{j=1}^L j^{2H} = nS_0$, the oracle constant is

$$\begin{aligned} C_{\text{oracle}} &= \mathrm{RMSE}(\hat{H}^*)\, \mathcal{I}_{n,L}(H)^{1/2} \\ &= \sqrt{\frac{S_0^2}{S_0 S_2 - S_1^2}} \\ &\approx 2.219. \end{aligned}$$

This corrects an arithmetic error in an earlier draft, where S_0 was misreported and the resulting constant was stated as 3.22. With the correct weighted sums, both the theoretical oracle constant and the simulated FWLS constant are approximately 2.22. The FWLS-to-oracle RMSE ratio is numerically indistinguishable from 1 in Tables 3 and 5.

Table 6: Sensitivity of the adequacy statistic to power-law misspecification.

Perturbation	Amplitude	Empirical rejection	Mean H	Mean Q
bump	0	0.05	0.60013	17.628216
bump	0.05	0.505	0.577	30.135298
bump	0.1	1	0.555295	70.732226
bump	0.2	1	0.515982	231.482208
sinusoid	0	0.045	0.600643	17.891493
sinusoid	0.05	1	0.604937	141.607201
sinusoid	0.1	1	0.608708	526.300995
sinusoid	0.2	1	0.612942	2044.392106
slope_shift	0	0.045	0.60003	18.34098
slope_shift	0.05	1	0.63834	70.382624
slope_shift	0.1	1	0.677072	231.136132
slope_shift	0.2	1	0.75676	866.936853

Table 7: Sensitivity of the adequacy test to alternative noise laws.

Noise law	Empirical rejection	Mean H	Mean Q
gaussian	0.06	0.5999	18.153075
bounded	0.005	0.599698	17.972076
student	0.06	0.599419	17.516054

C Auxiliary Lemmas

Lemma C.1 (KL divergence for Gaussian variance perturbations)**.** *For $\kappa^2 \le v$,*

$$D_{\mathrm{KL}}\big(\mathcal{N}(0,\kappa^2+v)\|\mathcal{N}(0,v)\big) = \frac{1}{2}\left[\frac{\kappa^2}{v} - \log\left(1+\frac{\kappa^2}{v}\right)\right] \le \frac{\kappa^4}{4v^2}.$$

Proof. Standard formula; the upper bound is the second-order Taylor remainder of $-\log(1+x)$ at $x = \kappa^2/v$.

Lemma C.2 (Pilot OLS rate in log-scale regression)**.** *In the pilot OLS regression of $Y_j = \alpha + Hx_j + U_j$ with $U_j \overset{iid}{\sim} \mathcal{N}(0,v)$ (ignoring heteroscedasticity), the OLS estimator $\hat{H}_{\mathrm{pil}}$ satisfies*

$$\hat{H}_{\mathrm{pil}} - H = \frac{\sum_j (x_j-\bar{x})U_j}{\sum_j (x_j-\bar{x})^2} \sim \mathcal{N}\left(0, \frac{v}{\sum_j (x_j-\bar{x})^2}\right).$$

For L fixed and $v = \sigma_0^2 n^{-1} j^{-2H}$ heteroscedastic, $\hat{H}_{\mathrm{pil}} - H = O_p(n^{-1/2})$.

Proof. Follows from the Gauss-Markov structure of OLS and the boundedness of $v_j \le \sigma_0^2/n$ (since $j \ge 1$, $j^{2H} \ge 1$).

References

Rainer Dahlhaus. Fitting time series models to nonstationary processes. *The Annals of Statistics*, 25(1):1–37, 1997. doi: 10.1214/aos/1034276620.

Eustasio del Barrio, Alberto Gonzalez-Sanz, Jean-Michel Loubes, and Jonathan Niles-Weed. An improved central limit theorem and fast convergence rates for entropic transportation costs. *SIAM Journal on Mathematics of Data Science*, 5(3):639–669, 2023. doi: 10.1137/22M149260X.

Nicolas Fournier and Arnaud Guillin. On the rate of convergence in Wasserstein distance of the empirical measure. *Probability Theory and Related Fields*, 162(3–4):707–738, 2015. doi: 10.1007/s00440-014-0583-7.

Ziv Goldfeld, Kengo Kato, Gabriel Rioux, and Ritwik Sadhu. Limit theorems for entropic optimal transport maps and the sinkhorn divergence. *Electronic Journal of Statistics*, 18(1):980–1041, 2024. doi: 10.1214/24-EJS2217.

Jan Nino G. Tinio, Mokhtar Z. Alaya, and Salim Bouzebda. Bounds in wasserstein distance for locally stationary functional time series. *CoRR*, abs/2504.06453, 2025. URL https://arxiv.org/abs/2504.06453.

Michael Vogt. Nonparametric regression for locally stationary time series. Technical Report CWP22/12, cemmap, 2012. URL http://www.cemmap.ac.uk/wps/cwp221212.pdf.